

2026 - Group 1 Preliminary Exam - Aptitude

Answer Key | Q45-Q198, 19 questions

Q.No	Question	Correct Answer	Syllabus	Explanation
45	The dimension of a brick are 24 cm x 12 cm x 8 cm. How many such bricks will be required to build a wall of 20 m length, 48 cm breadth and 6 m height? (A) 22000 (B) 21000 (C) 23000 (D) 25000	D) 25000	Part B, Unit I - Aptitude	Convert everything to the same unit (cm): the wall measures 2000 cm (length) x 48 cm (breadth) x 600 cm (height), giving a volume of $2000 \times 48 \times 600 = 57,600,000$ cubic cm. Each brick has a volume of $24 \times 12 \times 8 = 2,304$ cubic cm. Dividing the wall's volume by one brick's volume gives $57,600,000 / 2,304 = 25,000$ bricks, so the wall requires 25,000 bricks.
46	A, B, C took a house on rent for one year for Rs.13,824. They remained together for 4 months and then C left the house. After 5 more months, B also left the house. Then how much rent should 'B' pay? (A) Rs. 3,456 (B) Rs. 7,872 (C) Rs. 4,416 (D) Rs. 1,536	C) Rs. 4,416	Part B, Unit I - Aptitude	The monthly rent for the whole house is $\text{Rs.}13,824 / 12 = \text{Rs.}1,152$. For the first 4 months all three (A, B, C) share the house, so each pays $1,152 / 3 = \text{Rs.}384$ per month, giving B a share of $4 \times 384 = \text{Rs.}1,536$ for this period. For the next 5 months (months 5 to 9) only A and B remain, so they split the rent equally: $1,152 / 2 = \text{Rs.}576$ each per month, giving B $5 \times 576 = \text{Rs.}2,880$. B then also leaves, so B pays nothing for the final 3 months. Adding B's shares: $1,536 + 2,880 = \text{Rs.}4,416$, which is B's total rent contribution.
47	The difference between the 13th and 12th Fibonacci number is (if the first two numbers are $F_1 = 1$, $F_2 = 1$). (A) 144 (B) 89 (C) 55 (D) 34	B) 89	Part B, Unit I - Aptitude	Building the Fibonacci sequence from $F_1=1$, $F_2=1$: 1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, 144, 233 - so $F_{11}=89$, $F_{12}=144$ and $F_{13}=233$. The difference between the 13th term (233) and the 12th term (144) is $233 - 144 = 89$, which by the defining Fibonacci property (each term equals the sum of the two before it) is always exactly equal to the 11th term itself - confirming 89 as the answer.
48	In an A.P, S_1 , S_2 , S_3 are sum of first 15, 30, 45 terms respectively. If $S_3 = 300$,	C) 50	Part B, Unit I - Aptitude	For an arithmetic progression, there is a standard identity linking sums taken over equal-sized blocks of terms: $S_3 =$

	$S_2 = 150$ find S_1. (A) 150 (B) 100 (C) 50 (D) 30			$3 \times (S_2 - S_1)$, because the sums of successive equal blocks of an AP themselves form an AP. Substituting the given values: $300 = 3 \times (150 - S_1)$, so $100 = 150 - S_1$, which gives $S_1 = 50$.
92	A and B together can complete a piece of work in 15 days and B alone can do it in 20 days. In how many days can A alone complete the work? (A) 70 (B) 80 (C) 40 (D) 60	D) 60	Part B, Unit I - Aptitude	If A and B together finish the work in 15 days, their combined one-day work rate is $1/15$. B alone finishes in 20 days, so B's one-day rate is $1/20$. A's one-day rate is therefore $(1/15 - 1/20) = (4/60 - 3/60) = 1/60$, meaning A alone would take 60 days to complete the entire work. This is a standard 'time and work' problem solved by adding/subtracting work rates (reciprocals of time taken), giving option (D).
93	10 persons begin to work together on a job but after some days 4 persons leave. As a result, the job which could have been completed in 40 days is completed in 50 days. How many days after the commencement of the work did the 4 persons leave? (A) 20 days (B) 30 days (C) 35 days (D) 25 days	D) 25 days	Part B, Unit I - Aptitude	If 10 persons alone would finish the job in 40 days, then one person's daily work rate is $1/(10 \times 40) = 1/400$ of the job. Let x be the number of days all 10 persons worked before 4 left; during those x days they complete $10x/400 = x/40$ of the work. For the remaining $(50 - x)$ days, only 6 persons work, completing $6(50-x)/400$ of the work. Since the total must equal 1 (the whole job): $x/40 + 6(50-x)/400 = 1$. Multiplying through by 400 gives $10x + 300 - 6x = 400$, so $4x = 100$ and $x = 25$. Hence the 4 persons left 25 days after the work began, option (D).
94	A sum of Rs. 1,550 was lent partly at 5% and the remaining amount at 8% simple interest. The total interest received after 3 years was Rs. 300. The ratio of money lent at 5% to that lent at 8% is ____. (A) 5 : 8 (B) 8 : 5 (C) 31 : 6 (D) 16 : 15	D) 16 : 15	Part B, Unit I - Aptitude	Let P be the amount lent at 5% and $(1550 - P)$ the amount lent at 8%, both for 3 years using simple interest ($SI = \text{Principal} \times \text{Rate} \times \text{Time} / 100$). The total interest equation is: $(P \times 5 \times 3)/100 + ((1550 - P) \times 8 \times 3)/100 = 300$. Multiplying through by 100 gives $15P + 24(1550 - P) = 30000$, i.e., $15P + 37200 - 24P = 30000$, so $-9P = -7200$ and $P = 800$. The amount at 8% is then $1550 - 800 = 750$. The ratio of the two principal amounts is $800 : 750$, which simplifies to $16 : 15$, matching option (D).

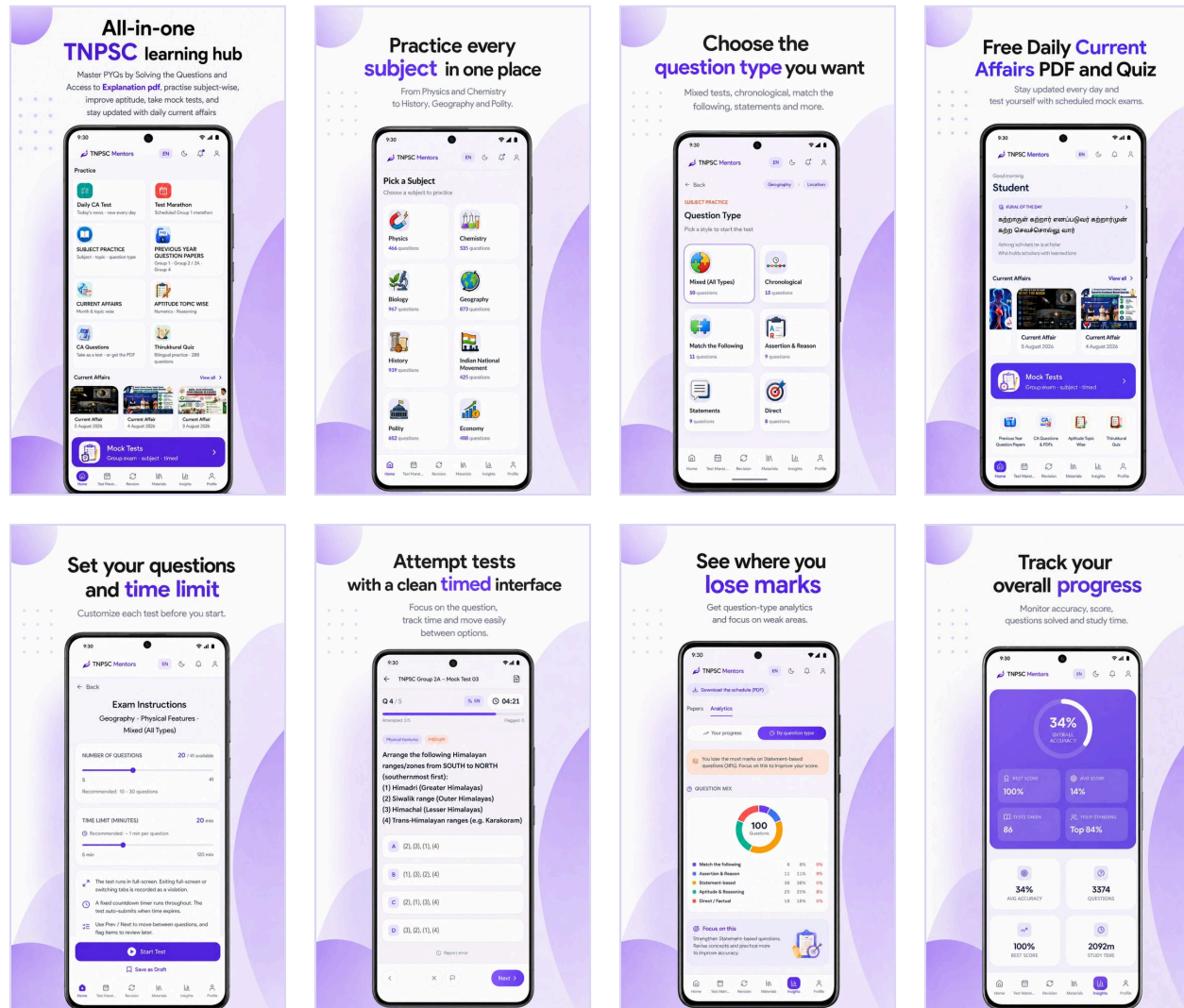
95	If a% of "a" is "a". Then what is b% of "a"? (A) a (B) b (C) c (D) None of these	B) b	Part B, Unit I - Aptitude	The condition 'a% of a is a' translates to $a \times (a/100) = a$. Dividing both sides by a (a is non-zero), $a/100 = 1$, so $a = 100$. Now b% of 'a' means b% of 100, which is $(b/100) \times 100 = b$. So the value of b% of a simplifies exactly to b, making option (B) correct. This is a classic percentage-algebra trick question where solving for the hidden value of 'a' (=100) is the key step.
96	If $x + y + z = 0$ then $x^3 + y^3 + z^3 + 3xyz$ is equal to _____. (A) 0 (B) $6xyz$ (C) $12xyz$ (D) xyz	B) $6xyz$	Part B, Unit I - Aptitude	There is a standard algebraic identity: $x^3 + y^3 + z^3 - 3xyz = (x + y + z)(x^2 + y^2 + z^2 - xy - yz - zx)$. When $x + y + z = 0$, the right-hand side becomes zero, so $x^3 + y^3 + z^3 - 3xyz = 0$, which means $x^3 + y^3 + z^3 = 3xyz$. Substituting this into the given expression: $x^3 + y^3 + z^3 + 3xyz = 3xyz + 3xyz = 6xyz$, confirming option (B).
97	A train Travels from Chennai to Delhi for 53 hours. If the train starts on Monday night 8.30 pm, find the day and time when the train would reach Delhi. (A) Wednesday 1.30 am (B) Wednesday 1.30 pm (C) Thursday 1.30 am (D) Thursday 1.30 pm	C) Thursday 1.30 am	Part B, Unit I - Aptitude	Starting from Monday 8:30 pm, add 53 hours in stages: adding 48 hours (2 full days) lands on Wednesday 8:30 pm. The remaining 5 hours ($53 - 48 = 5$) are then added to Wednesday 8:30 pm, giving Wednesday 8:30 pm + 5 hours = Thursday 1:30 am (since 8:30 pm + 3.5 hours reaches midnight, and 1.5 more hours reaches 1:30 am). So the train reaches Delhi at Thursday 1:30 am, option (C).
98	A store reduces the price of 1 product by 20%. How many % do you need to raise to the % to get the original price. (A) 30% (B) 27% (C) 25% (D) 20%	C) 25%	Part B, Unit I - Aptitude	Let the original price be 100. After a 20% reduction, the new (reduced) price is 80. To bring the price back up from 80 to 100, the required increase is $(100 - 80)/80 \times 100\% = 20/80 \times 100\% = 25\%$. Note that the percentage increase needed to reverse a decrease is always larger than the original percentage decrease, because the increase is calculated on the smaller, already-reduced base. Hence the store needs to raise the price by 25%, option (C).
144	Find the difference in compound interest and simple interest for principal 8000 and the rate 5% in 3 years. (A) Rs. 12 (B) Rs. 20	C) Rs. 61	Part B, Unit I - Aptitude	Simple Interest = $P \times R \times T / 100 = 8000 \times 5 \times 3 / 100 = \text{Rs. } 1,200$. Compound Interest = $P[(1 + R/100)^T - 1] = 8000 \times [(1.05)^3 - 1] = 8000 \times [1.157625 - 1] = 8000 \times 0.157625 = \text{Rs. } 1,261$. The difference is $CI - SI = 1,261 - 1,200 = \text{Rs. } 61$, matching option (C).

	(C) Rs. 61 (D) Rs. 51			
145	Divya bought Rs. 30,000 from a bank to buy a car at 20% simple interest. If she paid Rs. 18,000 as interest while clearing the loan, find the time for which the loan was given. (A) 6 years (B) 5 years (C) 4 years (D) 3 years	D) 3 years	Part B, Unit I - Aptitude	Using the Simple Interest formula, $SI = P \times R \times T / 100$. Substituting the known values: $18,000 = (30,000 \times 20 \times T) / 100 = 6,000 \times T$. Solving, $T = 18,000 / 6,000 = 3$ years, matching option (D).
146	HCF of $x^{(a+9)}$, $x^{(a+7)}$, $x^{(a+5)}$ is _____. (A) x^a (B) $x^{(a+9)}$ (C) $x^{(a+7)}$ (D) $x^{(a+5)}$	D) $x^{(a+5)}$	Part B, Unit I - Aptitude	For powers of the same base, the Highest Common Factor is the base raised to the smallest exponent among the given terms, because that power divides evenly into all the higher powers (e.g., $x^{(a+5)}$ divides $x^{(a+7)}$ and $x^{(a+9)}$ exactly, leaving x^2 and x^4 respectively). Among the exponents (a+9), (a+7) and (a+5), the smallest is (a+5), so the HCF is $x^{(a+5)}$, option (D).
147	The income of a person is increased by 20% and then decreased by 20%. Then the change in his income is _____. (A) 1% increase (B) 3% decrease (C) 5% increase (D) 4% decrease	D) 4% decrease	Part B, Unit I - Aptitude	When a value is increased by x% and then decreased by the same x%, the net percentage change is always a decrease of $x^2/100$ percent. Here $x = 20$, so the net change $= 20^2/100 = 400/100 = 4\%$ decrease. (Worked directly: let income = 100; after a 20% increase it becomes 120; after a 20% decrease of 120, it becomes $120 - 24 = 96$, which is a decrease of 4 from the original 100.) This matches option (D).
195	If side of a square is "x" meter and area of that square is x square meter, then which of the following is true? (A) It is not possible (B) It is true for all values of x (C) It is true for x = 1, 4, 9, 16 (D) It is true for x = 1	D) It is true for x = 1	Part B, Unit I - Aptitude	If the side of the square is x metres, its area is x^2 square metres by the standard formula for the area of a square. The problem states that this area equals x (not x^2 in general, but the numerical value x itself), so we need $x^2 = x$. Rearranging: $x^2 - x = 0$, i.e. $x(x - 1) = 0$, which gives $x = 0$ or $x = 1$. Since a square's side length cannot be zero, the only valid solution is $x = 1$ (at which point the side is 1 m and the area is also 1 sq. m, so the

				two numerically coincide). This equation is not satisfied for all x, nor specifically for the list 1, 4, 9, 16 (those are perfect squares of side lengths 1,2,3,4, which is irrelevant here), so option (D) is correct.
196	A sum of money triples itself in 12 years when the principal and compound interest are added. In how many years will it become nine times the sum? (A) 36 years (B) 24 years (C) 42 years (D) 72 years	B) 24 years	Part B, Unit I - Aptitude	Under compound interest, if a sum P grows to 3P in 12 years, then $P(1+r)^{12} = 3P$, so $(1+r)^{12} = 3$. We want to find time t such that the sum becomes 9 times, i.e. $(1+r)^t = 9$. Since $9 = 3^2 = [(1+r)^{12}]^2 = (1+r)^{24}$, we get t = 24. So the sum, which triples every 12 years under compound interest, triples again in the next 12 years ($3 \times 3 = 9$ times) after a total of 24 years. Hence option (B), 24 years, is correct.
197	If $(x + y) : (x - y) = 7 : 3$ then find the ratio $(x^3 + y^3) : (x^3 - y^3)$ (A) 240/315 (B) 33/21 (C) 233/188 (D) 133/117	D) 133/117	Part B, Unit I - Aptitude	Given $(x+y):(x-y) = 7:3$, let $x + y = 7k$ and $x - y = 3k$ for some constant k. Adding these two equations: $2x = 10k$, so $x = 5k$; subtracting: $2y = 4k$, so $y = 2k$. Now compute $x^3 = (5k)^3 = 125k^3$ and $y^3 = (2k)^3 = 8k^3$. Then $x^3 + y^3 = 125k^3 + 8k^3 = 133k^3$, and $x^3 - y^3 = 125k^3 - 8k^3 = 117k^3$. The ratio $(x^3+y^3):(x^3-y^3) = 133k^3 : 117k^3 = 133:117$. Hence option (D) is correct.
198	If LCM of a and b is 100 and $a + b = 45$. Find the difference between a and b. (A) 55 (B) 5 (C) 25 (D) 15	B) 5	Part B, Unit I - Aptitude	We need two numbers a and b whose LCM is 100 and whose sum is 45. Testing factor pairs of 100 that add to 45: 20 and 25 add up to 45 ($20 + 25 = 45$), and their LCM is indeed 100 (since $20 = 2^2 \times 5$, $25 = 5^2$, and LCM takes the highest powers: $2^2 \times 5^2 = 100$). No other factor pair of 100 sums to 45, so $a = 20$ and $b = 25$ (in either order). The difference between them is $25 - 20 = 5$. Hence option (B) is correct.

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